Stem-Cell Evidence: Benefit, Risk, Uncertainty

LAUNCH · Science · 40 minutes

I can analyse simulated stem-cell evidence by calculating outcomes and identifying benefits, risks and limitations.

Prepare before pupils arrive

Setup: Print the simulated table with SIMULATED TRAINING DATA in its title. Prepare calculators and percentage frames.

Safety: Never describe the fictional numbers as a study, trial or treatment result without the word simulated.

Equipment: Simulated data sheet, calculator, percentage frame, benefit-risk-uncertainty grid and pencil.

Safety: State repeatedly that the dataset is fictional and not medical advice. Keep all cases impersonal and allow a no-discussion calculation route.

Fallback: A pupil selects numerator, denominator and operation while an adult operates the calculator and scribes exact analysis.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Discuss potential benefits and risks associated with the use of stem cells in medicine.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Define benefit in this scientific context. Define risk without making it sound certain.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	benefit — possible helpful outcome risk — chance of harm uncertainty — what is not yet secure Label the table SIMULATED TRAINING DATA. Treatment: 12 of 20 improved; comparison: 6 of 20 improved.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Add risk: four of 20 treatment participants and one of 20 comparison participants had the defined adverse event. Add uncertainty: each group is small, follow-up is short and the simulation gives no evidence about long-term outcomes.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 5 min	Place each evidence statement, then use the frozen set to make one cautious analysis. READ DATA CALCULATE CLASSIFY LIMIT ANALYSE Use the numbered cards in your pupil pack.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 12 min	Analyse the simulated table: calculate four percentages, classify benefit and risk, and state two uncertainties. Record: Visible percentage working plus a benefit-risk-uncertainty analysis that does not overclaim.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	State one benefit finding and one uncertainty from the simulated data. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: 4 ÷ 20 × 100 = 20%

Yes. Divide the frequency by the total number in that group, then multiply by 100.

Misconception repair

Repaired. The simulation suggests a possible benefit and a possible risk, but its size and duration limit the conclusion.

Guided checkpoint

Yes. Divide the frequency by the total number in that group, then multiply by 100. Use frequency ÷ total × 100. The total is 20.

Improvement frequency

BENEFIT EVIDENCE. Benefit evidence: the simulated treatment group recorded more improvement. This card reports a potentially helpful outcome.

Adverse-event frequency

RISK EVIDENCE. Risk evidence: the adverse event was recorded more often in the simulated treatment group. This card reports a potentially harmful outcome.

Short follow-up

UNCERTAINTY / LIMIT. Uncertainty: eight weeks cannot show long-term benefit or harm. Ask what the study duration cannot reveal.

Model limitation

The numbers are invented for classroom reasoning, the sample is small and no real treatment conclusion can be drawn from them.

Independent evidence

Visible percentage working plus a benefit-risk-uncertainty analysis that does not overclaim.

Supported: Complete two percentage calculations with the formula shown, then match one benefit, risk and limit. Numbers are pre-positioned: frequency ÷ 20 × 100. Use icons and exact scribing if needed.

Standard: Calculate all four percentages and write a three-part analysis. Use: The data suggest __. However __. We cannot conclude __ because __.

Stretch: Calculate all outcomes, compare groups and evaluate how sample size and follow-up affect confidence. Do not add new biology; deepen the evidence judgement and propose one useful next-data need.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre’s authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner’s practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W11L2_Benefit_Risk_Uncertainty_Explore.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.